

PROJECT REPORT

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CS -510

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PROJECT TITLE:

FIRE & GAS LEAKAGE

DETECTION SYSTEM

Introduction

The Fire Detection Control Center is a software-based smart monitoring system developed using Python and Streamlit. The main purpose of this project is to monitor fire-related dangers such as gas leakage, high temperature, and flame detection in real time.

This project simulates an intelligent fire and gas detection environment where sensor values are generated automatically and displayed on a modern dashboard. The system continuously checks the sensor readings and identifies whether the environment is SAFE, WARNING, or DANGER.

The project also stores all monitoring data inside a SQLite database and can send alert notifications through email when dangerous conditions are detected.

This system is useful for:

- Homes
- Industries
- Laboratories
- Offices
- Warehouses
- Smart buildings

The project is completely software-based and does not require physical hardware sensors. Random values are generated to simulate real sensor readings.

Objective

The main objectives of this project are:

- To develop a smart fire and gas monitoring system.
- To monitor gas level, temperature, and flame detection.
- To provide real-time monitoring using a dashboard.

- To classify conditions into SAFE, WARNING, and DANGER.
- To store sensor data in a database for future analysis.
- To generate automatic alerts during dangerous situations.
- To create a user-friendly and attractive monitoring interface.
- To simulate IoT-based fire detection using software.

Methodology

The methodology of this project consists of multiple steps:

1. System Initialization

When the application starts:

- Streamlit initializes the dashboard.
- SQLite database table is created automatically.
- User interface and styling are loaded.

2. Sensor Data Generation

The system generates random values for:

- Gas level
- Temperature
- Flame detection

Example:

- Gas Range = 100 to 1000
- Temperature Range = 20°C to 100°C
- Flame Detection = YES or NO

These values simulate real sensors.

3. Threshold Checking

The system compares generated values with threshold values.

Default Thresholds:

- Gas Threshold = 700
- Temperature Threshold = 70°C

Conditions:

- If gas or temperature exceeds threshold → DANGER
- If gas level approaches threshold → WARNING
- Otherwise → SAFE

4. Dashboard Monitoring

The dashboard displays:

- Gas level
- Temperature
- Flame status
- Alert notifications
- Graphs and analytics

5. Database Storage

Each sensor reading is stored inside SQLite database with:

- Time
- Gas value
- Temperature
- Flame status
- System status

6. Alert System

When dangerous conditions occur:

- The system can send email alerts.
- Emergency warning message is generated.

Functionality

The project contains multiple functionalities.

1. Real-Time Monitoring

The system continuously updates sensor values in real time.

Features:

- Live monitoring
- Automatic refresh
- Dynamic dashboard

2. Gas Detection

The system monitors gas concentration levels.

If gas level becomes dangerous:

- Warning message appears
- Danger alert activates

3. Temperature Detection

The system checks environmental temperature.

If temperature exceeds the threshold:

- Fire risk is detected
- System status changes to DANGER

4. Flame Detection

The system randomly simulates flame presence.

If flame is detected:

- Emergency warning is displayed
- System enters danger mode

5. Alert Classification

The project classifies conditions into:

SAFE

- Normal environment
- Green alert box

WARNING

- Moderate risk
- Yellow alert box

DANGER

- Critical condition
- Red emergency alert box

6. Data Visualization

The system displays:

- Area charts
- Sensor analytics
- Historical data trends

7. Database Management

SQLite database stores all sensor records.

Benefits:

- Data tracking
- History analysis
- Future reporting

8. Email Alert System

The system includes an email alert feature using SMTP.

When danger occurs:

- Email notification can be sent to the user.

Tools and Devices

Software Tools Used

Tool / Technology	Purpose
Python	Main programming language
Streamlit	Dashboard and web interface
SQLite	Database management
Pandas	Data handling and analysis
SMTP Library	Sending email alerts
Random Module	Simulating sensor values
Datetime Module	Time tracking

Devices / Components

Since this is a software-based simulation project, no physical hardware is used.

Virtual Components:

- Gas Sensor Simulation
- Temperature Sensor Simulation

- Flame Sensor Simulation
- Alert System
- Database System

Optional Real Hardware (Future Enhancement):

- MQ-2 Gas Sensor
- Flame Sensor
- DHT11 Temperature Sensor
- ESP32 or Arduino
- Buzzer
- Wi-Fi Module

Project Structure

Files Included

1. app.py

Main application file.

Functions:

- Dashboard creation
- Sensor simulation
- Status detection
- Data visualization

2. database.py

Handles SQLite database operations.

Functions:

- Create table
- Insert sensor data

3. alerts.py

Handles email notifications.

Functions:

- Send danger alerts

- SMTP email communication

4. fire_data.db

SQLite database file.

Stores:

- Sensor readings
- Alerts
- Time records

Working Principle

Step-by-Step Working

1. Application starts.
2. Database table is created.
3. Random sensor values are generated.
4. Threshold comparison is performed.
5. System status is identified.
6. Dashboard updates in real time.
7. Data is stored in SQLite database.
8. Alerts are generated if danger occurs.

User Interface Description

The dashboard contains:

1. Sidebar Settings

User can control:

- Refresh speed
- Gas threshold
- Temperature threshold

2. Sensor Cards

Displays:

- Gas Level
- Temperature
- Flame Status

3. Alert Box

Shows:

- SAFE message
- WARNING message
- DANGER message

4. Analytics Graph

Displays live sensor data trends.

Advantages

1. Easy to use
2. Real-time monitoring
3. Attractive dashboard interface
4. Data storage support
5. Email alert system
6. Software simulation without hardware
7. Useful for learning IoT concepts
8. Expandable for future development

Limitations

1. Uses simulated sensor values instead of real hardware.
2. Requires internet for email alerts.

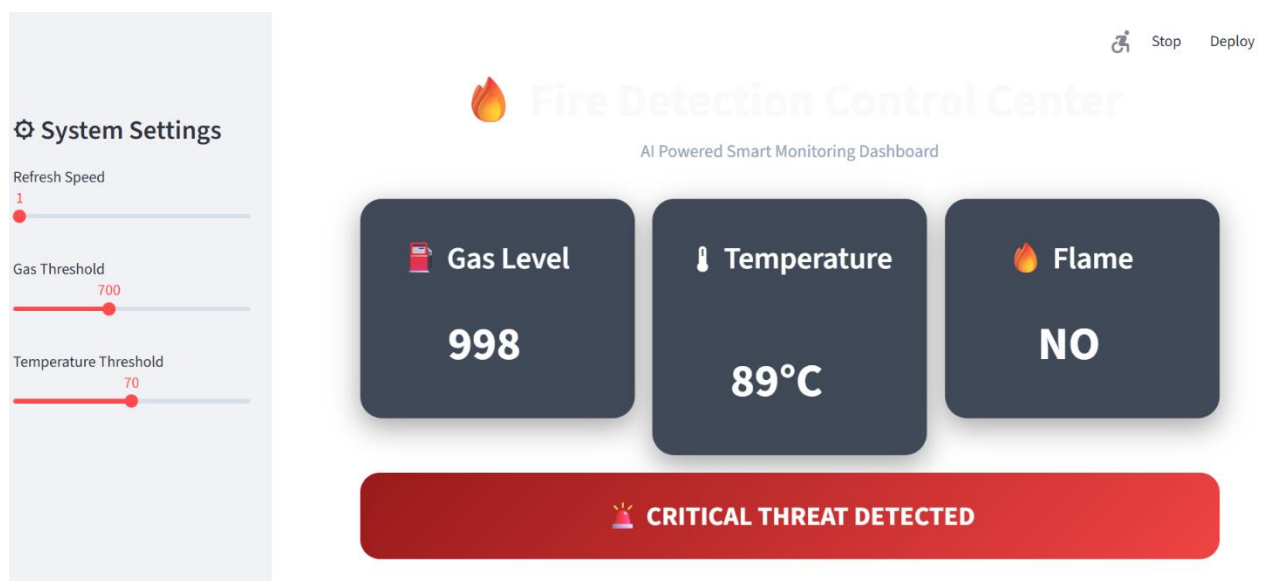
3. Not connected to physical fire safety equipment.
4. Accuracy depends on simulated data.

Future Enhancements

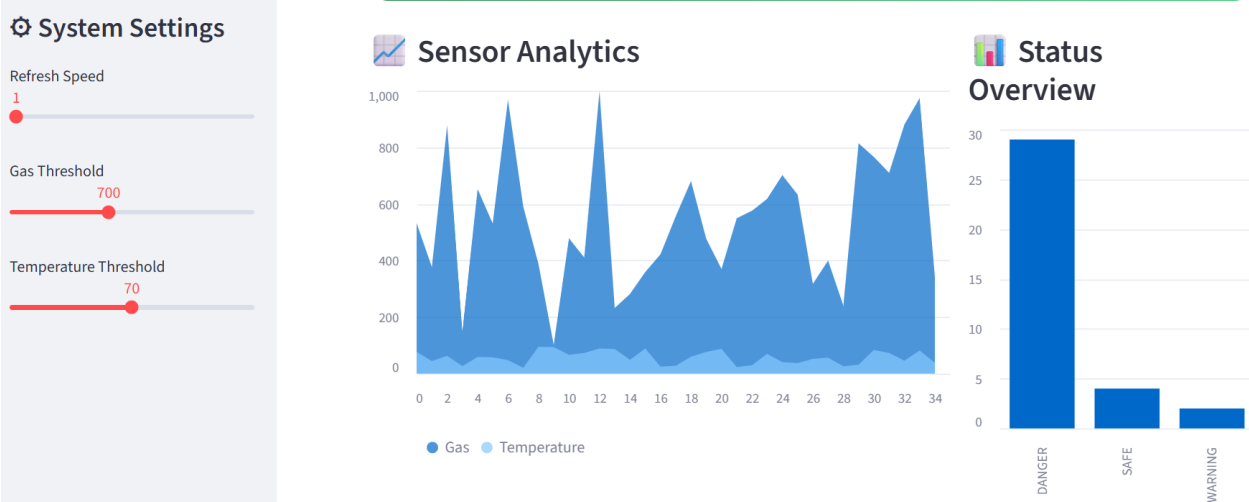
Future improvements can include:

1. Integration with real IoT sensors.
2. Mobile application support.
3. SMS alert system.
4. AI-based fire prediction.
5. Cloud database integration.
6. CCTV monitoring integration.
7. Voice alarm system.
8. Web deployment for remote access.

Main Dashboard:



Analytics Graph:



Database Records:

System Settings

Refresh Speed

Gas Threshold

Temperature Threshold

Event Logs

	Time	Gas	Temperature	Flame	Status
0	20:22:10		532	77 NO	DANGER
1	20:22:15		377	45 YES	DANGER
2	20:22:17		877	63 YES	DANGER
3	20:22:18		155	27 NO	SAFE
4	20:22:19		653	59 NO	WARNING
5	20:22:21		531	58 YES	DANGER
6	20:22:22		968	48 YES	DANGER
7	20:22:23		591	21 YES	DANGER
8	20:22:25		388	95 NO	DANGER
9	20:22:26		102	94 YES	DANGER

DANGER Alert:



bilalsa... 4 days ago
to me ▾



ALERT DETECTED!

Gas Level: 968
Temperature: 72°C

Show quoted text



bilalsa... 4 days ago
to me ▾



ALERT DETECTED!

Gas Level: 537
Temperature: 29°C